MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROPOLYMER INSULATED, EXTRUDED TYPE
NICKEL-COATED HIGH STRENGTH COPPER ALLOY CONDUCTOR, 600-VOLT

This specification is approved by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in effect of Specification MIL-W-22759.

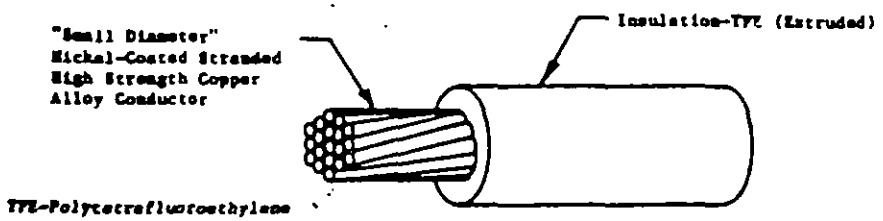


TABLE I. CONSTRUCTION DETAILS

Part No. 1/	Wire size	Stranding (Number of strands X AWG gage of strands)	Diameter of stranded conductor (inches)		Finished wire		
			(min)	(max)	Resistance at 20°C (68°F) (ohms/1000 ft) (max)	Diameter (inches)	Weight (lbs/1000 ft) (max)
M22759/23-28-*	28	7 X 36	.014	0.016	79.0	.033 ±.002	1.34
M22759/23-26-*	26	19 X 38	.018	0.020	49.4	.038 ±.002	1.92
M22759/23-24-*	24	19 X 36	.023	0.025	30.1	.043 ±.002	2.63
M22759/23-22-*	22	19 X 34	.029	0.031	18.6	.049 ±.002	3.73
M22759/23-20-*	20	19 X 32	.037	0.040	11.4	.058 ±.002	5.44

1/ PART NO.: The asterisks in the part number column, Tables I and II, shall be replaced by color code designators in accordance with MIL-STD-681. Examples:- Size 20, white - M22759/23-20-9; white with orange stripe - M22759/23-20-93.

FSC 6145

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TABLE II. PERFORMANCE DETAILS

Part No	Abrasion resistance (Procedure II)				Bend testing			
					Mandrel diameter (inches) (±3%)		Test load (lbs) (±3%)	
	Resistance (inches of tape) (min) (initial condition)	Weight support bracket	Weight (lbs)	Tension load (lbs)	Life cycle (oven & bend tests) 1/	Cold bend test	Life cycle (oven & bend tests) 1/	Cold bend test
M22759/23-28-*	12	A	.50	1.0	.125	.250	.50	
M22759/23-26-*	18	A	.50	1.0	.125	.250	.50	
M22759/23-24-*	18	A	.50	1.0	.125	.250	.50	
M22759/23-22-*	24	A	.50	1.0	.250	.375	.75	
M22759/23-20-*	24	A	.50	1.0	.250	.375	.75	

1/ Also for bend tests after immersion.

WIRE RATINGS AND ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 260°C (500°F) max conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level
ABRASION RESISTANCE AFTER IMMERSION: No requirement
ACID RESISTANCE: Required
Dielectric test, 2200 volts (rms), 60Hz
BLOCKING: 260 ±2°C (500 ±3.6°F)
COLOR: In accordance with MIL-STD-104, Class 1; white preferred
COLOR STRIPING OR BANDING DURABILITY: 250 cycles (500 strokes) (min), 500 grams weight
DIELECTRIC TEST AFTER IMMERSION: 2200 volts (rms), 60Hz
FLAMMABILITY: Post-flame dielectric test not required
HUMIDITY RESISTANCE: No requirement

IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
IMPULSE DIELECTRIC TEST: 8.0 kilovolts (peak), 100% test
INSULATION RESISTANCE: 50,000 megohms for 1000 ft (min)
LIFE CYCLE: Oven temperature 275 ±2°C (527 ±3.6°F)
Dielectric test, 2200 volts (rms), 60Hz
LOW TEMPERATURE (COLD BEND):
Bend temperature: -65 ±2°C (-85 ±3.6°F)
Dielectric test, 2200 volts (rms), 60Hz
SHRINKAGE: 0.03 inch max at 290 ±2°C (554 ±3.6°F)
SMOKE: 290°C (554°F)
SPARK TEST OF PRIMARY INSULATION: Not required
SURFACE RESISTANCE: 500 megohm-inches (min), initial and final readings
THERMAL SHOCK: Oven temperature, 260 ±2°C (500 ±3.6°F)
Max change in measurement, 0.060 inch
WICKING: No requirement
WIRE LENGTH REQUIREMENTS: Schedule A
WRAP TEST: "Wrap back" test required; no cracking
Oven temperature, 313 ±2°C (595.4 ±3.6°F)

Preparing activity:
Navy - AS
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